BreezeSwap

Pooled Underwriting for On-Chain Weather Risk

Clearing parametric weather exposure without a matched counterparty

Protocol Whitepaper

Version 1.0

August 2026

Status. The protocol described here is implemented in Solidity and deployed to the Flare Coston2 testnet. Every quantitative result in this paper was produced by running the repository's own test and simulation suites, and every figure is derived either from that output or from a thirty-year observational record. Section 12 states plainly what has not been built, what has not been audited, and which live addresses do not yet carry the capital structure described in Section 6. This document is a technical description of a protocol. It is not an offer, a solicitation, or investment advice.

Contents

Abstract

Weather moves a large share of global economic output, yet the instruments that transfer weather risk reach almost nobody. Exchange-traded weather futures settle against thirteen United States cities and require a clearing relationship. Parametric insurance protocols pay automatically but sell cover in one direction only and must find an underwriter before a policy exists. Both designs share a single structural dependency: a hedger cannot trade until somebody willing to take the other side arrives at an acceptable price. For weather that dependency is close to fatal, because hedging demand is directional by construction. Farmers in a region want drought cover in the same month, and none of them wants to sell it.

BreezeSwap removes the matching requirement rather than trying to solve it. A shared underwriting pool stands as the counterparty to every product the protocol offers, so a single buyer and a single pool constitute a market. Prices come from a thirty-year observational record rather than from the balance of arriving orders, which means the pool can quote before any order flow exists. This paper sets out the economics of that design.

We make four contributions. First, we formalise why counterparty matching fails under directional demand and show that the fraction of hedging demand that clears in a two-sided venue falls to zero as demand becomes one-sided, while a pooled venue clears every order until capital binds. Second, we derive the protocol's capital multiplier from its published parameters and show that one unit of senior capital supports up to 2.4 units of worst-case directional notional, or 1.5 units when a single market is the binding constraint. Third, we show that a subordinated tranche only enlarges a protocol when the capital is genuinely additive; Monte Carlo runs across five seeds show that carving a junior tranche out of a fixed capital base makes senior capital worse on one of five paths, whereas adding it protects senior on all five. Fourth, we show that pricing weather at even odds is not a simplification but a transfer. On the thirty-year Tokyo record, monthly August rainfall exceeded 40 mm in 28 of 30 years, so a contract collateralised 50/50 hands the buyer of the exceedance side an expected edge of 43 percentage points per unit of cover.

We also report a result that cuts against the protocol's current configuration. Sweeping the reserve coverage ratio across three seeds and 900 actions each under the most aggressive funding parameters the protocol permits, the shipped setting of 50% produced one payout shortfall while 75% produced none, at a cost of 0.5 percentage points of additional trade rejection. Section 8 presents the sweep in full and argues that the parameter should move.

Keywords: weather derivatives, parametric insurance, automated market making, pooled underwriting, tranched capital, adverse selection, correlated risk.

1 Introduction

1.1 The gap between weather risk and weather markets

Weather is one of the largest uninsured exposures in the economy and one of the least tradeable. A rice farmer outside Tokyo carries the entire variance of the monsoon on their own balance sheet. A ski operator in Nagano carries a season's revenue against snowfall they cannot influence and cannot hedge. A solar operator in Dubai has a revenue line that is a direct function of irradiance. Each of these is a well-defined, measurable, financially material exposure, and for each of them the available instruments are either unavailable at their size, unavailable in their location, or slow enough to arrive after the loss has already forced a decision.

Three markets exist and each fails a different part of the population. Exchange-traded weather derivatives at CME settle against heating and cooling degree days for thirteen United States cities [6]. That coverage reflects where the clearing infrastructure and the institutional counterparties are, not where weather risk sits. Traditional indemnity insurance covers the right people but pays on an adjuster's assessment, which introduces a claims cycle measured in months at precisely the moment a policyholder needs liquidity. Parametric insurance solves the claims cycle by settling on a measured index rather than an assessed loss, and blockchain-based parametric protocols have taken that idea further by making settlement automatic [7, 3]. What they have not solved is supply. A parametric policy still requires an underwriter to commit capital against a specific peril before the policy can be written, and the underwriter is a scarce, slow-moving, relationship-mediated resource.

1.2 Why the on-chain attempts stall

The obvious on-chain construction is a two-sided market: post a threshold, let one side take the over and the other the under, escrow both deposits, and pay out against an oracle. This is a prediction market with a weather index attached, and it is the design most readily built. It has two defects that only appear at deployment.

The first defect is that it silently prices every outcome as a coin flip. If both sides post equal collateral, the market has asserted that the threshold is breached half the time. Weather thresholds are not fifty-fifty. On the thirty-year record we use throughout this paper, monthly August rainfall in Tokyo exceeded 40 mm in 28 of 30 years. A contract on that threshold with equal collateral is not a market. It is a standing instruction to transfer money from whoever does not know the climatology to whoever does.

The second defect is that a two-sided market requires two sides. Weather hedging demand is structurally one-directional within a region and season: everyone exposed to drought wants the same protection at the same time, and the natural seller is not another farmer but a capital provider with no weather exposure at all. Section 3 makes this precise. The short version is that as demand becomes one-sided, the fraction of hedgers who find a counterparty falls toward zero, and the market's failure mode is not bad pricing but an empty book.

1.3 The approach

BreezeSwap replaces counterparty matching with pooled underwriting. Capital providers deposit into a shared vault, and that vault takes the other side of every product the protocol lists. A buyer who wants drought cover does not wait for a seller; they pay a premium computed from the historical breach frequency of their strike, and the vault reserves the payout. A trader who wants continuous exposure to a rainfall index does not wait for an opposing trader; they trade against a virtual automated market maker whose solvency is backed by the same vault.

This is not a novel idea in isolation. Pooled counterparty models exist in perpetual futures [15, 9], and mutualised underwriting capital exists in on-chain insurance [13]. The contribution here is in what pooling weather risk specifically requires, which turns out to be considerably more than pointing a vault at a weather oracle. Weather risk is severely correlated within a region and season, so a pool that treats each policy as an independent draw is holding one large bet while its accounting reports a diversified book. Weather risk is also forecastable, so a buyer who can see a two-week forecast holds a conditional probability that the pool's unconditional historical price does not reflect. Weather indices are seasonal, so a single annual probability misprices every individual month. Each of these has a specific mechanism in the protocol, and each of them is the difference between a design that works in a demonstration and one that survives contact with informed buyers.

1.4 Contributions

1. **A clearing model for directional risk** (Section 3). We formalise the matched fraction of hedging demand under a two-sided venue and show it decays as $(1 - p)/p$ in the directional share p, versus unit clearing for a pooled venue up to a capital bound.
2. **A derived capital multiplier** (Section 6). From the protocol's utilisation cap, concentration cap, coverage ratio, and junior backing share, we derive closed-form bounds on the notional a given quantity of senior capital supports, and state where each bound binds.
3. **A measured, not asserted, capital structure** (Sections 7 and 8). The loss waterfall and the reserve coverage ratio are calibrated by simulation, and we report the results including the one that argues against the shipped parameter.
4. **Climatological pricing with an enforced sample floor** (Section 5). Strike probabilities carry the sample size behind them, are keyed per calendar month, and gate market creation and deposits on-chain rather than appearing only in a user interface.

2 Background and related work

2.1 Exchange-traded weather derivatives

CME lists futures and options on heating degree days and cooling degree days for thirteen United States cities, with monthly and seasonal structures [6]. The academic literature on valuing these instruments is mature. Alaton, Djehiche and Stillberger model temperature as a mean-reverting process with seasonal drift and derive prices for degree-day options [1]. Cao and Wei estimate the market price of weather risk and find it is not zero, which matters because it means a weather contract cannot be priced by the historical distribution alone; the risk premium is a real component of the fair price [5]. Jewson and Brix provide the standard practitioner treatment [11].

BreezeSwap adopts the finding directly. Its premium is the historical expected loss plus an explicit, floored risk load, and the floor exists because underwriting at historical fair value is a losing business once variance is priced.

2.2 Basis risk and index design

An index-settled contract does not pay the holder's actual loss; it pays a function of a measured index. The gap between the two is basis risk, and it is the central design constraint of parametric cover. Woodard and Garcia show that weather hedges lose a substantial fraction of their effectiveness as the distance between the measurement station and the exposure grows, and that aggregating exposure upward recovers some of it [18]. BreezeSwap does not solve basis risk and does not claim to. It makes the index explicit, settles on the same statistic the premium was priced from, and leaves the choice of station and threshold to the buyer.

2.3 Correlated risk and the failure of private crop insurance

The most directly relevant result in the insurance literature is Miranda and Glauber's analysis of why private crop insurance markets fail without reinsurance [12]. Their finding is that systemic weather effects induce high correlation across farm-level yields, which defeats pooling: they estimate that United States crop insurer portfolios are twenty to fifty times riskier than they would be if yields were independent across farms.

This is the single most important external result for a protocol proposing to pool weather risk on-chain. A vault that writes a hundred drought policies across one region and season has not written a hundred independent risks; it has written one risk a hundred times. The protocol's per-peril exposure caps (Section 9) exist because of this result, and they are the mechanism that stops the accounting from reporting diversification that does not exist.

2.4 Adverse selection and forecast skill

Rothschild and Stiglitz established that when buyers hold information the insurer does not, the pooling equilibrium is unstable and the informed buyers select against the pool [16]. For weather this is not a hypothetical asymmetry, it is a scheduled one. Numerical weather prediction has genuine skill to roughly two weeks, and that horizon has been extending by about one day per decade for forty years [4]. A buyer who can see a usable forecast of the period being covered holds a conditional probability; the pool, pricing off thirty years of unconditional history, does not. The protocol's response is a minimum lead time between purchase and the start of the covered period, which is the on-chain form of the sales closing date that crop insurance has used for decades.

2.5 On-chain parametric insurance

Etherisc provides a generic parametric insurance framework with automatic payouts [7]. Arbol, working with dClimate data infrastructure, writes parametric weather cover at institutional scale [3]. Both are competent at what they do and both are insurance in the strict sense: cover is unidirectional, held to expiry, and requires an underwriter to be found first. Nexus Mutual pioneered mutualised on-chain underwriting capital with a minimum capital requirement governing how much cover the pool may write [13]; BreezeSwap's underwriting utilisation ceiling plays the same role.

2.6 Automated market making and pooled perpetuals

Constant-product market makers price without an order book [17], and their properties as price oracles and their vulnerability to manipulation are well characterised [2]. Perpetual Protocol introduced the virtual AMM, in which the reserves are accounting fictions used purely for price discovery and no real assets sit in the curve [15]. GMX popularised the pooled-counterparty model in which liquidity providers take the other side of trader flow and are compensated with fees and trader losses [9].

A virtual AMM produces synthetic exposure, not liquidity. When traders are net profitable, there is no counterparty whose losses fund those profits, and without a backstop the market pays winners from the collateral of traders who have not yet closed. BreezeSwap's liquidity vault is that backstop, and Section 6 describes the reserve model that makes the obligation bounded rather than open-ended.

2.7 Where BreezeSwap sits

Table 1 compares the venues on the five properties a hedger actually cares about. The row that matters is not settlement, where the parametric protocols are already good, but whether a counterparty has to exist before a hedge can be placed.

3 The liquidity problem

3.1 Directional demand is structural, not transient

In most derivatives markets the two sides of a contract are populated by participants with genuinely opposing views or opposing exposures. An airline is short jet fuel and a producer is long it. That symmetry is what makes an order book viable.

	Settlement	Both sides	Exit early	Priced off history	Retail reach
CME weather futures	Exchange	Yes	Yes	Yes	13 US cities
Arbol / dClimate	Parametric	Cover only	No	Yes	Institutional
Etherisc	Parametric	Cover only	No	Per product	Yes
Indemnity crop insurance	Adjuster	No	No	Actuarial	Yes, slow
Naive on-chain binary pool	Oracle threshold	Yes	Partial	No	Yes
BreezeSwap	Parametric	All products	Yes	Enforced on-chain	Yes

Table 1: Positioning against existing venues. The column that separates BreezeSwap from the parametric protocols is not settlement quality, which is comparable, but whether a counterparty must be found before a hedger can transact.

Weather has no such symmetry within a region and season. Consider drought cover for a single growing region in a single month. Everyone with an agricultural exposure in that region is on the same side, because they share the same weather. The natural seller is not another participant with the opposite weather exposure, because that participant would have to be someone who profits from drought in the same place at the same time, and such participants are rare. The natural seller is a capital provider with no weather exposure at all, who is selling variance rather than expressing a view.

A two-sided venue cannot manufacture that capital provider. It can only wait for one.

3.2 A clearing model

Let λ be the arrival rate of interest in a market and let $p \in [0.5, 1]$ be the fraction of that interest wanting the same direction, which for weather is the hedging direction. In a venue that clears by matching opposing interest, the volume that can clear in a period is bounded by twice the smaller side, so the fraction of hedging demand that finds a counterparty is

$$M_{\text{p2p}}(p) = \frac{\min(p, 1-p)}{p} = \frac{1-p}{p} \qquad \text{for } p \geq \tfrac{1}{2}.$$

In a venue where a capital pool takes the other side of every order, the matched fraction is

$$M_{\text{pool}}(p) = \begin{cases} 1 & \text{while reserved capital is below the utilisation cap,} \\ \dfrac{K}{\lambda p} & \text{once capacity } K \text{ binds,} \end{cases}$$

where K is the capacity derived in Section 6. The distinction matters because the two failure modes are different in kind. The pooled venue fails by running out of capital, which is a solvable problem: capital responds to yield. The two-sided venue fails by running out of opinions, which is not, because the missing participant is missing for a structural reason.

3.3 Why isolating markets is the wrong response

The standard DeFi answer to risk concentration is to isolate pools, so that a failure in one market cannot reach another. That answer is correct when markets are genuinely independent. Weather markets are not. Rainfall in Tokyo and rainfall in Osaka in the same week can be the same storm system. Isolation in that setting produces the appearance of diversification while the underlying peril is shared, which is precisely the failure Miranda and Glauber identify in crop insurance [12].

BreezeSwap therefore shares capital across markets and bounds the shared exposure explicitly, rather than partitioning capital and hoping the partitions are meaningful. Section 9 describes the mechanism and reports what happens without it.

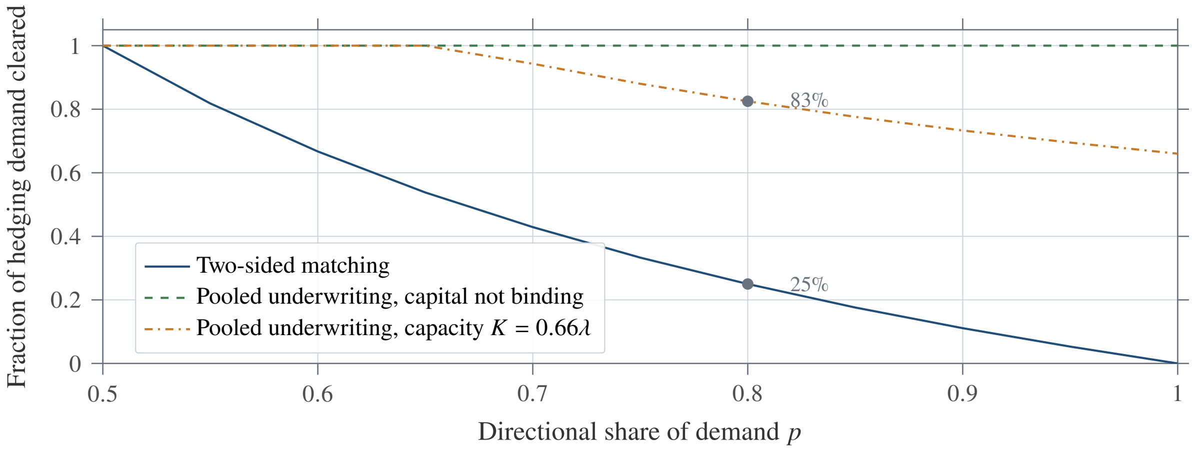

Figure 1: Clearing under directional demand. At $p = 0.8$, which is a mild assumption for regional weather hedging, a two-sided venue clears one hedger in four. A pooled venue clears every hedger until reserved capital reaches the utilisation cap, and then degrades smoothly rather than emptying. The capacity-bound curve is drawn for illustrative K; the derivation of K from protocol parameters is in Section 6.

4 Protocol design

4.1 Three products, one balance sheet

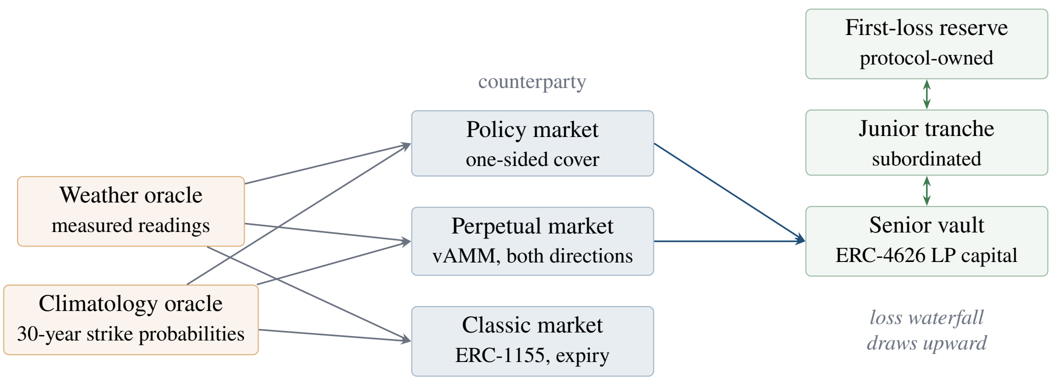

Figure 2: Product surface and capital stack. The classic market clears peer to peer and does not consume vault capital; the policy and perpetual markets clear against the vault. All three price against the same climatology feed, so a strike cannot be quoted in one product at odds the protocol refuses in another.

Policy market. A buyer pays a premium and receives a promise: if the aggregated index over the covered period breaches a strike, the payout is transferred. This is the Arbol-equivalent product, and it clears with one buyer because the vault underwrites it.

Perpetual market. A trader takes long or short exposure to a weather index against a virtual constant-product curve, with leverage capped at 3, a 10% maintenance margin, and funding that pulls the mark toward the oracle index. Positions can be closed at any time, which is the property none of the insurance products offer.

Classic market. An expiry-dated pool where both sides deposit collateral and receive ERC-1155 position tokens that can be transferred or sold before settlement. Payoffs may be binary, linear, or capped, and the calculator guarantees that the long and short payouts sum exactly to the notional.

4.2 The vAMM and why it needs a backstop

The perpetual market prices against synthetic reserves (x, y) satisfying $x \cdot y = k$, with mark price $P = x/y$. No real assets sit in the curve. Opening a long increases x and decreases y; the trader's position size is the change in y.

This construction gives price discovery without liquidity providers posting into a curve, which is exactly what a market with no natural two-sided flow needs. What it does not give is a source of funds. In a matched book, one trader's profit is another's loss and the market is self-funding. In a skewed book, the winning side's profit has no source, and without a backstop the market pays it out of the collateral posted by traders who have not yet closed, so the last trader to exit absorbs the shortfall.

> **The core obligation**
>
> A vAMM manufactures exposure, not capital. Every design that uses one either bounds the skew it will accept, or backs the skew with real capital, or quietly makes its last exiting trader the insurer of everyone before them. BreezeSwap does the first two and publishes the bound.

Funding is the first line of defence and works on incentives: when the mark sits above the oracle index, longs pay shorts, which makes the crowded side expensive to hold. Funding is zero-sum only across matched open interest, so it cannot close a structural skew on its own. The skew cap and the capital reserve do that.

4.3 Three defects found by measurement

Three implementation defects are worth recording because each was found by instrumenting the system rather than by reading it, and each was invisible to tests that only checked whether a call succeeded.

1. **Oracle scale mismatch.** Mark price was 1e18-scaled and the oracle reading was 1e6-scaled. Nothing reconciled the two, so every funding calculation saw a deviation of order 10^{16} basis points and clamped to the maximum rate in the same direction on every call. Funding was not pulling the mark toward the index; it was a fixed maximum levy on longs that no market condition could change. The market looked functional throughout.

2. **Settlement measuring a different quantity from pricing.** The climatology script sums daily precipitation into a monthly total and counts how often that total breached the strike, so a 40 mm threshold means 40 mm across the month. Settlement compared a single reading at expiry against that threshold. One day of rainfall is almost always below a monthly total, so drought cover triggered on nearly every policy while being charged the monthly-total probability. The pricing was correct and the settlement was measuring something else.

3. **Unbounded iteration on the payout path.** The function computing open-position obligations looped over every position the market had ever created. That loop only grew, so the sweep of realised trader losses to liquidity providers was guaranteed to exceed the block gas limit eventually and stay there. It is a liveness failure on a timer, and no test that opens a realistic number of positions would find it.

Each now carries a regression test. The reason for listing them is that they are the class of defect that separates a protocol that demonstrates well from one that holds capital safely, and none of them is visible in an architecture diagram.

5 Pricing weather

5.1 Even odds is a transfer, not a simplification

The premium for a binary weather claim is

$$\pi = L \cdot \hat{p} \cdot (1 + \lambda),$$

where L is the payout limit, $\hat{p}$ is the historical breach frequency for the specific region, variable, direction, threshold, and calendar month, and λ is the risk load. The load is floored at 5% and defaults to 30%, with a further surcharge that scales with how committed the pool is, capped at 50% in total.

The floor exists because underwriting at historical fair value is a losing business. The expected profit is zero by construction, and the realised outcome has variance, so a pool charging exactly $L\hat{p}$ has a positive probability of ruin and a zero expected reward for accepting it. Cao and Wei's empirical finding that the market price of weather risk is not zero is the same statement from the other direction [5].

Now compare against a contract collateralised at even odds, which is what a naive two-sided pool implements. That contract charges $0.5L$ regardless of $\hat{p}$. The wedge is

$$\Delta(\hat{p}) = L\,(\hat{p} - 0.5)\,,$$

which is an expected transfer per unit of cover from the uninformed side to the informed one.

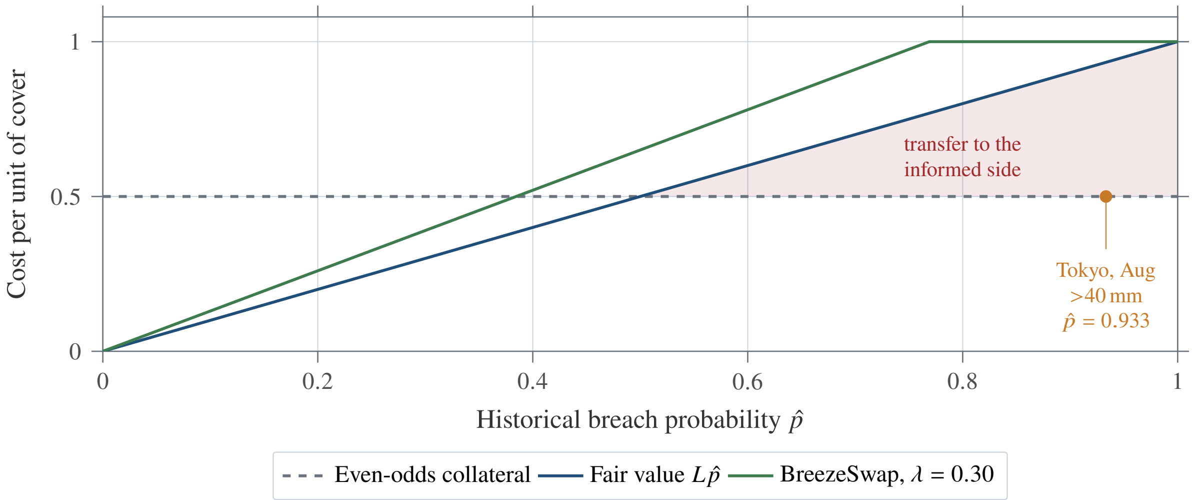

Figure 3: Pricing a weather claim at even odds transfers value in proportion to how far the true frequency sits from one half. On the Tokyo August 40 mm strike, where the threshold was breached in 28 of 30 years, the transfer is 43.3 percentage points of the payout limit per contract. The BreezeSwap premium tracks fair value with a floored load and is capped strictly below the payout, so a buyer always retains something to win.

5.2 The empirical record

Strike probabilities are computed off-chain from the Open-Meteo archive [14], which reanalyses observational data into a gridded historical record built on ERA5 [10]. The seeded dataset used throughout this paper covers 1996 to 2025, thirty complete years, across five regions and nine thresholds for each calendar month and each direction, producing 1,080 priced strikes. Each on-chain entry carries the sample size behind it, and the contract refuses any entry derived from fewer than ten years.

Figure 4 shows why a single global threshold is meaningless. In August, the probability that monthly rainfall exceeds 80 mm is 96.7% in Singapore, 90.0% in Seoul, 70.0% in Tokyo, 16.7% in London, and 0% in Dubai. The same nominal contract is a near certainty in one place and an impossibility in another.

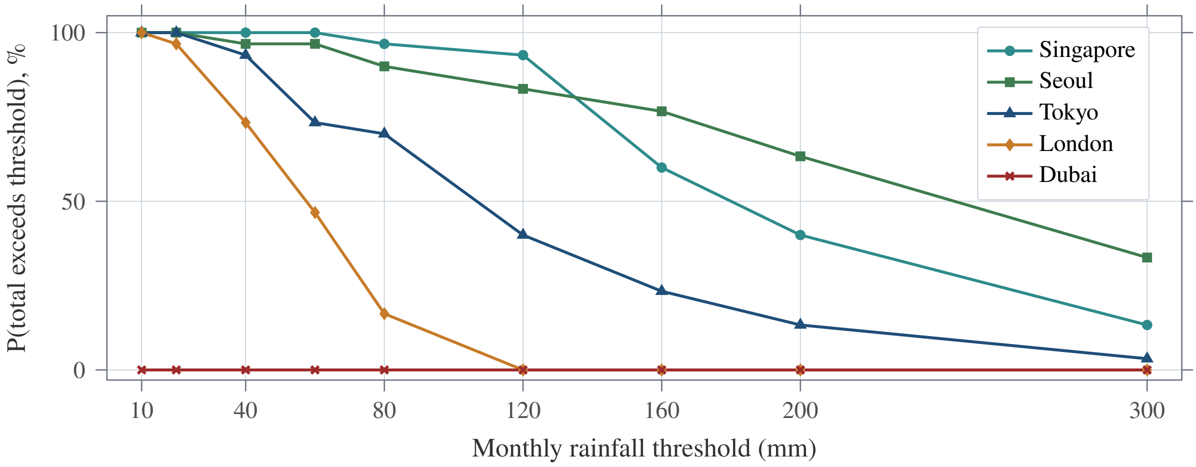

Figure 4: Empirical August exceedance curves, 1996 to 2025, from the seeded Open-Meteo archive. Thirty observations per point. Dubai records no August month above 10 mm in the sample, which is a correct answer and a reason the protocol will not list a market whose strike has no informative history.

5.3 Seasonality forces per-month pricing

A single annual probability misprices every month. Figure 5 shows the probability that Tokyo monthly rainfall exceeds 80 mm by calendar month: it runs from 23.3% in February to 96.7% in June. A protocol pricing the annual average of 71.1% would sell February cover at roughly three times fair value and June cover at roughly three quarters of it, and the second error is the expensive one because it is the one informed buyers will take.

The strike key therefore includes the calendar month, and the month is derived on-chain from the block timestamp rather than supplied as a parameter, so a market cannot be pointed at a season whose climatology happens to flatter it.

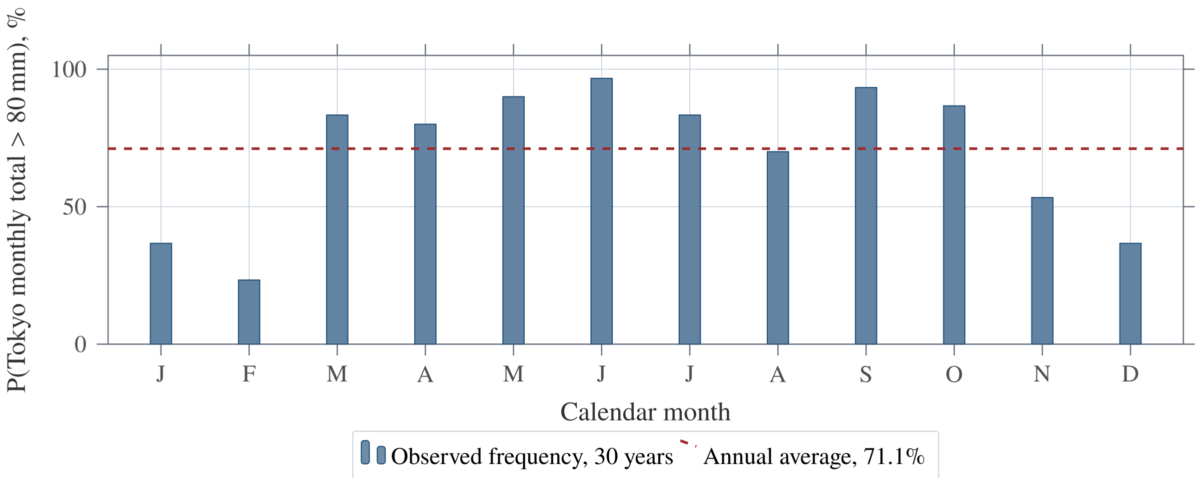

Figure 5: Seasonality in a single strike. A protocol pricing off the annual average would overcharge winter cover by a factor of three and undercharge June cover by a quarter.

5.4 The distribution is skewed, so the mean is not the price

Figure 6 plots the thirty August totals for Tokyo in ascending order. The mean is 117.5 mm and the median is 104.7 mm; the standard deviation is 69.4 mm and the maximum, 315.5 mm, is 4.5 times the minimum. The distribution has a long right tail, which is the normal shape for precipitation and the reason the protocol prices from empirical quantiles rather than fitting a symmetric distribution and reading off a mean.

It is also the reason the risk load is not cosmetic. An underwriter selling exceedance cover on this series is short the right tail, and the tail is where the capital goes.

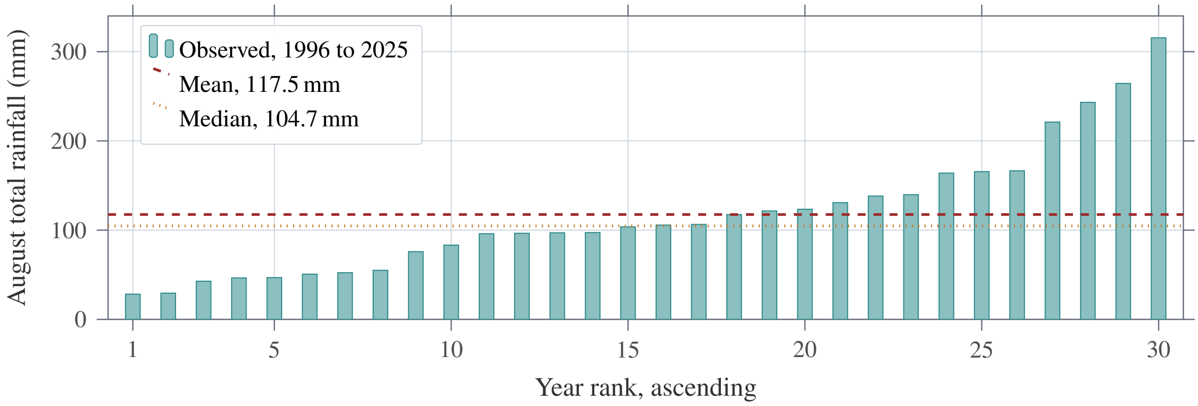

Figure 6: Tokyo August monthly rainfall totals, sorted. Mean above median, standard deviation 69.4 mm, and a maximum 4.5 times the minimum. Precipitation is right-skewed, so an underwriter selling exceedance cover is short a tail rather than short a symmetric bet.

5.5 Three gates, and what each refuses

Pricing that lives only in a user interface is decoration. Each product therefore carries an on-chain gate that can refuse a transaction outright, and Table 2 states what each one turns away.

Product	Gate	What it refuses
Classic, binary	Fair-odds band on the long share of the pool	A deposit that pushes the implied odds further from the historical frequency once already outside tolerance. A deposit that moves the pool toward fair odds is always allowed, so a mispriced pool stays repairable.
Perpetual	Opening mark within a band of the climatological expected level for the month	Market creation at a mark unrelated to the climate the index tracks, which would misprice every subsequent trade.
Policy	Premium computed from the strike probability, load floored at 5%	Selling cover at or below expected loss, and any quote derived from fewer than ten years of record.

Table 2: Pricing gates by product. Linear and capped classic payoffs are deliberately left unpriced: they require the loss distribution rather than a single quantile of it, and the protocol flags them as unpriced on-chain rather than attaching an authoritative-looking wrong number.

5.6 Adverse selection and the sale window

Historical frequency is an unconditional probability. A buyer who can see a forecast of the period being covered holds a conditional probability instead, and will buy only when the conditional exceeds the price. That is not bad luck for the pool; it is systematic bleed, and it is the mechanism Rothschild and Stiglitz describe [16].

Numerical weather prediction carries real skill to roughly two weeks, with marginal skill extending to about six [4]. The protocol therefore refuses to sell cover whose covered period begins sooner than a configured lead time, defaulting to 45 days and floored at 21 days so the parameter cannot be tuned back inside the forecast horizon. This is the same instrument as a crop insurance sales closing date, expressed as a contract condition rather than an underwriting policy.

5.7 Settlement must measure what pricing measured

If the premium was derived from the frequency with which a monthly total breached a threshold, then settlement must compare a monthly total against that threshold. Each policy therefore records an aggregation at purchase, summing readings for rainfall totals and averaging them for temperature means, and settlement collapses the whole covered period using the recorded statistic rather than reading a single value at expiry.

Two further conditions follow. Readings are sampled on an exact grid and a reading whose timestamp does not match is treated as absent rather than substituted, because an oracle that falls back to the most recent value would silently contribute the previous day twice into a sum. And a settlement requires at least 93% of expected samples to be present, matching the tolerance the climatology script applies when it discards incomplete months. The tolerance is tight in this direction specifically: a missing sample under-counts a total, which biases drought cover toward paying out, so the error favours the buyer and it is the capital providers who need the bound.

6 Capital economics

This is the section that determines whether the protocol works with less liquidity than its competitors or merely claims to.

6.1 What the protocol locks, and why gross

For policy cover the vault reserves the gross payout limit, not the limit net of premium. Reserving net looks correct, since the premium is paid in and the two together appear to cover the promise. They do not, for a timing reason. Premium becomes recognised liquidity provider value on a vesting schedule with no relation to the policy's term, while the obligation runs to expiry. A claim then arrives short by exactly the premium, and the effect grows with the strike probability: on a strike with a 70% breach frequency the premium is most of the payout, so almost nothing would be reserved and the cover would be largely notional. Cover is collateralised to its full limit, as a catastrophe bond is [8].

For perpetual markets the reserve requirement is a fraction of worst-case directional notional rather than the whole of it, because a perpetual position's loss is bounded by price movement rather than by a binary payout. Two choices in that definition are load-bearing.

Worst case, not current skew. The requirement is sized on $\max(N_{\text{long}}, N_{\text{short}})$, not on $|N_{\text{long}} - N_{\text{short}}|$. A balanced book of 100k against 100k has zero skew and would reserve nothing, but the moment one side closes the imbalance is 100k, and by then the capital providers whose funds would have backed it have already been free to withdraw. Sizing on the larger side prices the exposure the book can reach without a single new position opening.

Preventive, not reactive. Capacity is derived from backing that already exists and an opening trade is refused before any token moves, so an under-reserved state is unrepresentable rather than merely detectable. Exits are never gated by capacity, which matters because a book pushed over its cap by a tightened parameter or a capital provider withdrawal must stay repairable.

6.2 Deriving the capital multiplier

Let S be senior vault assets, J junior tranche assets, u the aggregate utilisation cap, c the single-market concentration cap, σ the coverage ratio applied to worst-case notional, and β the maximum share of counted backing that junior capital may supply. The shipped parameters are

$$u = 0.80, \quad c = 0.50, \quad \sigma = 0.50, \quad \beta = 0.3333.$$

Counted backing is senior assets plus credited junior assets, where credit is capped so that junior can never exceed β of the total. Solving $c_J \le \beta(S + c_J)$ gives

$$B \;=\; S + \min\!\left(J,\; \frac{\beta}{1-\beta}\,S\right) \;\le\; \frac{S}{1-\beta} \;=\; 1.50\,S.$$

The reserve a market must hold against worst-case notional N is σN. Summed across markets the utilisation cap gives

$$\sigma \sum_i N_i \;\le\; uB \quad\Longrightarrow\quad \sum_i N_i \;\le\; \frac{u}{\sigma} B \;=\; 1.6\,B \;\le\; 2.40\,S,$$

and for any single market the concentration cap binds first:

$$N_{\text{single}} \;\le\; \frac{c}{\sigma} B \;=\; 1.0\,B \;\le\; 1.50\,S.$$

For policy cover the binding constraints are an aggregate outstanding-payout cap of 60% of senior assets and a per-peril cap of 20%, so a pool of size S supports $0.6S$ of outstanding limit spread across at least three distinct region-months.

> **The multiplier, stated plainly**
>
> One unit of senior capital supports up to 2.40 units of worst-case directional notional across the perpetual markets, or 1.50 units if a single market is the whole book, and 0.60 units of fully collateralised policy limit outstanding at any moment. The perpetual figures require the junior tranche to be subscribed to its counted share; without it they are 1.60 and 1.00 respectively.

6.3 What this means for capital providers

The gross expected return on senior capital from policy underwriting, at full use of the aggregate cap, is

$$r \;=\; \underbrace{0.60}_{\text{aggregate cap}} \;\times\; \underbrace{\frac{365}{T}}_{\text{turnover}} \times \hat{p} \times \lambda,$$

where T is the cover term in days. At the protocol's binding term limit of 62 days under daily sampling and the default 30% load, this is $r \approx 1.06\,\hat{p}$ per year.

The number that makes this interesting is not its level but its correlation. Weather risk is close to uncorrelated with crypto asset returns, so a vault whose drawdowns are driven by rainfall does not draw down at the same moment as the rest of a DeFi portfolio. That is an unusual property in on-chain yield and it is the reason capital has a reason to arrive at all.

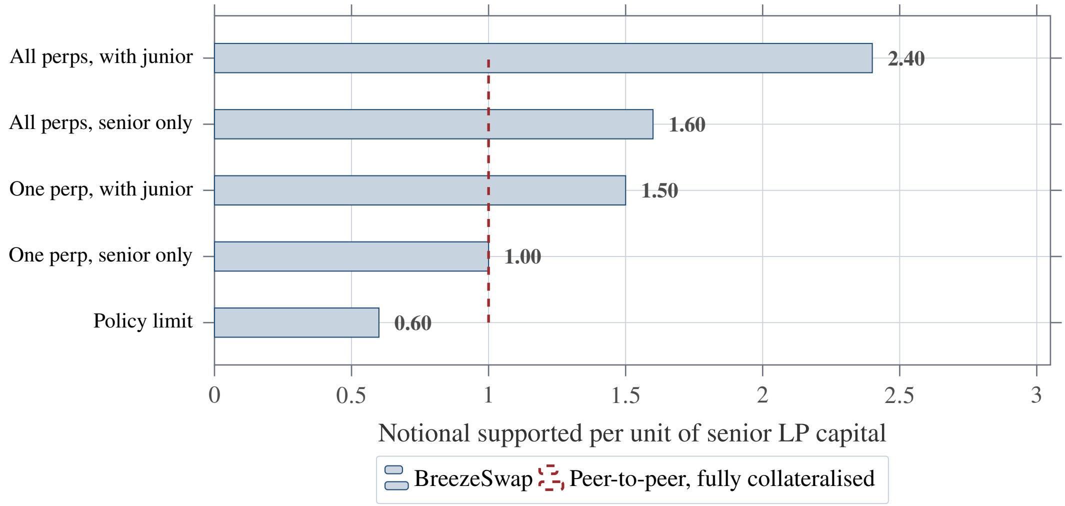

Figure 7: Capital multiplier by constraint, derived from the shipped parameters. The dashed line is a fully collateralised peer-to-peer contract, which supports exactly one unit of notional per unit of capital and additionally requires a matched counterparty to exist. Note the honest reading: policy cover is *less* capital-efficient than a bilateral contract, at 0.60. Its advantage is that it clears at all.

Strike breach probability $\hat{p}$	0.10	0.20	0.30
Expected premium per unit of limit	0.130	0.260	0.390
Expected loss per unit of limit	0.100	0.200	0.300
Expected margin per unit of limit	0.030	0.060	0.090
Gross annual return on senior capital	10.6%	21.2%	31.8%

Table 3: Gross expected return on senior capital from policy underwriting, at full use of the aggregate exposure cap, 62-day terms, and the default 30% risk load. These are expectations before variance, before protocol fees, and before any allowance for the periods in which the book is not at the cap. A pool writing high-probability strikes earns more in expectation and pays claims far more often, so the variance of the realised path rises with $\hat{p}$ as well.

7 The loss waterfall

7.1 Structure

A loss that a market cannot fund is drawn in a fixed order, and each tier's contribution is emitted so the ordering is verifiable from outside the contract.

Two design points deserve emphasis. Tier 2 is a layer, not a bucket: attachment and exhaustion points are expressed as fractions of total backing, and the width becomes a per-period aggregate limit, so junior's exposure is bounded in time and therefore priceable. And tier 1 has its own dedicated balance rather than sharing one with the liquidation backstop, because sharing let the vault drain the reserve that liquidation depends on.

7.2 Tranching redistributes loss, it does not reduce it

The claim that a subordinated tranche protects senior capital is comparative, so testing it requires a comparative experiment: the same trader flow, from the same seed, run through capital structures holding the same total risk capital. A structure that is safer because it is simply larger has demonstrated nothing about tranching.

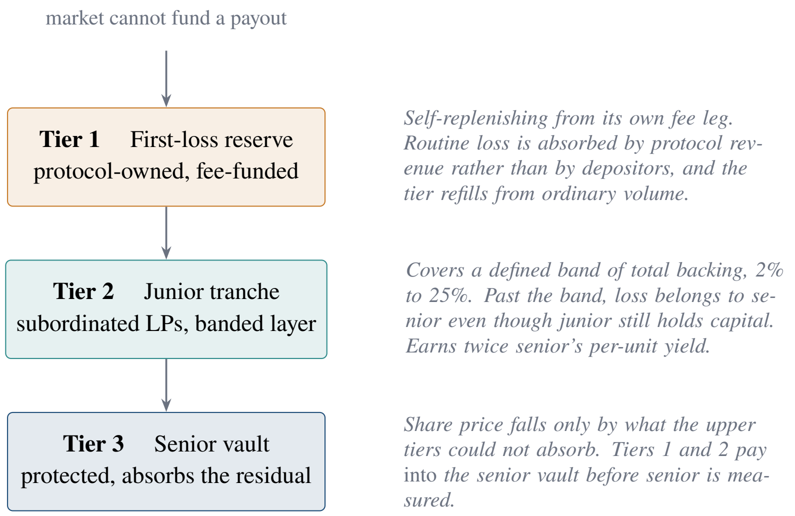

Figure 8: The three-tier loss waterfall. Both upper tiers are optional; with neither configured the structure collapses to single-pool behaviour and the depositor interface is unchanged.

The simulation runs four arms over 40 traders, 6 capital providers, and 250 actions per seed across 5 seeds, with 400,000 units of total capital:

- **Arm A**: single tranche, 400k senior. The pre-waterfall baseline.
- **Arm C**: 300k senior plus 100k junior, carved out of the same total.
- **Arm D**: 400k senior plus 100k *additional* junior.

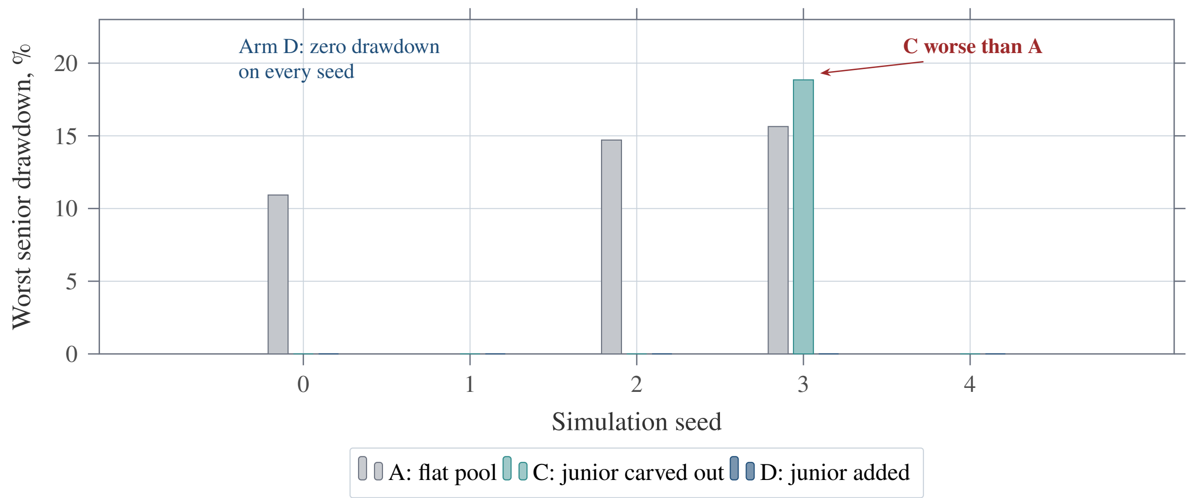

Figure 9: Worst senior drawdown by arm and seed, measured as the largest fall in senior share price from its starting value of 1.0. Arm C beats the flat pool on four seeds and loses on seed 3, where a senior tranche 25% thinner moves further per unit of residual loss once junior is exhausted. Arm D, where junior capital is genuinely additional, never draws down at all. Mean drawdown: A 8.26%, C 3.77%, D 0.00%.

Seed 3 is the whole result. Arm C's worst senior share price of 0.8115 is worse than the flat pool's 0.8436, because once the junior layer is exhausted the residual falls on a senior tranche that is 25% smaller. Tranching moved the loss; it did not remove it.

Why junior capital must be additive, and how that is enforced

A vault cannot control where a depositor's money came from, so "additive" cannot be enforced at the point of deposit. What it can control is whether junior capital *buys capacity*. Counting junior toward backing only up to $\beta = 1/3$ of the total means a protocol that funds itself by moving senior depositors into the junior tranche gets no extra capacity for it, while one that attracts genuinely new subordinated capital does. Excess junior capital is not rejected and not trapped: it still absorbs loss, and it stays withdrawable, because capital that backs no capacity cannot be holding any up.

A separate arm measured whether tier 1 alone helps. Across five seeds the dedicated first-loss reserve improved the mean worst senior share price from 0.9174 to 0.9300 and was drawn on three of five seeds, with no seed made worse. When the same tier shared a balance with the liquidation backstop rather than holding its own, it measured worse for senior than having no tier 1 at all on two of five seeds.

8 Calibrating the reserve

8.1 The trade-off

The coverage ratio σ sets how much of worst-case directional notional must be backed by capital. Raising it makes shortfalls less likely and rejects more trades. Lowering it does the reverse, up to a point: a pool that drains rejects everything, so under-reserving is capable of being worse on both axes at once.

The sweep runs three seeds of 900 actions each under the most aggressive funding parameters the protocol permits. That preset is chosen deliberately. Under production funding parameters every ratio including 30% survives, so that regime cannot discriminate; it shows the protocol is comfortable, not that 30% is safe. Insurance capital is sized on the tail.

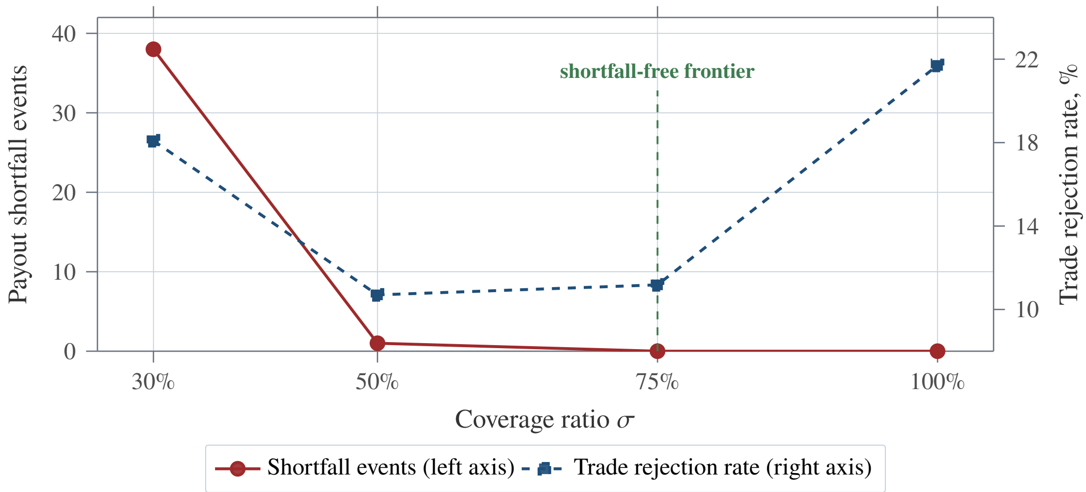

Figure 10: Reserve coverage sweep, 3 seeds by 900 actions under the aggressive funding preset. At 30% the pool both fails to pay traders and rejects more trades than at 50%, because a drained pool refuses everything. The shipped setting of 50% produced one shortfall event with a worst severity of 30.9% of the amount owed. At 75% shortfalls disappear at a cost of 0.48 percentage points of additional rejection.

8.2 The result argues against the shipped parameter

We report this rather than round it away. The protocol currently ships $\sigma = 0.50$, and under the aggressive funding preset that setting produced one shortfall event across 2,700 simulated actions, with a worst severity of 30.9% of the amount a trader was owed. Setting $\sigma = 0.75$ eliminated it and cost 0.48 percentage points of additional trade rejection.

Coverage ratio	30%	**50% (shipped)**	75%	100%
Shortfall events	38	**1**	0	0
Worst shortfall, % of amount owed	100.0	**30.9**	0	0
Mean trade rejection rate	18.09%	**10.70%**	11.18%	21.70%
Seeds ending near-drained	2 of 3	**0**	0	0

Table 4: Reserve calibration sweep, measured. Under-reserving is dominated: 30% is worse on both the solvency axis and the liquidity axis. The interesting comparison is 50% against 75%, where 0.48 percentage points of extra rejection buys the elimination of the remaining shortfall.

The reserve calibration also has a known hole: 40% was not tested, so the failure boundary is only located somewhere between 30% and 50%. Section 12 carries both points forward as open items rather than resolving them here.

8.3 Scarcity is priced, not only rationed

A hard capacity cap is a cliff. Below it every trade costs the same; at it, nothing gets through. That is the wrong shape for a scarce resource, because it neither discourages the marginal trade that fills the last of the book nor pays capital providers more for the tail exposure they take as the book fills.

The protocol therefore charges a surcharge that scales linearly with capacity utilisation, from zero on an empty book to 0.40% of collateral on a full one, applied only to flow that raises worst-case exposure. A trade that grows the smaller side consumes no scarce capacity and pays nothing extra. The proceeds are retained in the market, which puts its balance above open-position obligations, so the existing surplus sweep remits them to capital providers.

The surcharge does not replace the cap. The cap is a solvency bound derived from capital that exists; the surcharge makes the approach to it expensive so the cliff binds less often. Removing the cap and relying on price would mean accepting exposure the capital cannot support at any fee.

9 Correlated exposure

Two rainfall markets on regions that see the same storm system are not two risks. Each market's own capacity bounds it against the shared pool, but nothing bounded the correlated set, so both could fill their allowance against a single weather event.

Measured without a group cap, two correlated rainfall markets committed 599,400 against a 1,000,000 pool, a 50% overshoot of what any single peril's cap permits. The registry buckets markets by declared peril and takes the tighter of a market's own capacity and what its peril group leaves it.

Exposure is pulled from registered markets rather than mirrored in registry state. Push accounting would require every open, close, and liquidation to update a copy of state that already exists, and any missed path leaves the copy wrong in a way nothing detects. Pulling cannot desynchronise because there is nothing to synchronise. The loop is bounded by a governance-set maximum, which is what keeps a correctness mechanism from becoming a gas trap.

Three properties hold throughout. A cap refuses a trade only when it actually raises worst-case exposure, so a balancing trade is accepted even while the book is over its cap. Exits are never gated. And a registry that reverts degrades to the market's own capacity rather than blocking trading, because a correctness improvement must not become a liveness failure.

10 Protecting capital providers from each other

Three mechanisms exist because a pooled vault creates opportunities to extract value from other depositors that a bilateral contract does not.

Limit	Setting	Binds on
Aggregate vault utilisation	80% of backing	All reservations, both product lines
Ceiling on that cap	90%	Immutable; no admin action raises it
Single-market concentration	50% of backing	One market's reservation
Policy aggregate outstanding	60% of senior assets	Total payout limit written
Policy per-peril outstanding	20% of senior assets	One region-month bucket
Underwriting halt	70% utilisation	New policy sales only
Perp open-interest skew	25% of initial reserve	Unmatched open interest
Junior share of counted backing	33.33%	Capacity credit, not loss absorption

Table 5: Exposure limits. Each is a separate bound and the tightest one governs. The underwriting halt sits deliberately below the vault's own reservation ceiling so the pool keeps a solvency buffer rather than writing cover down to the last unit, which is the role a minimum capital requirement plays in a mutual [13].

Premium vests, it does not land. Premium is earned across the life of the risk it pays for. Crediting it at the moment of sale would let a depositor arrive one block before a sale, capture the whole premium through the jump in share price, and leave the next block having carried none of the exposure. Recognition is linear over a configurable period, defaulting to seven days and floored at one, and the vesting deadline is stored explicitly so that repeated trivial deposits cannot restart the clock on premium already part-way through vesting.

Exit is delayed, never blocked. Losses hit share price the moment a claim settles, while profit is recognised gradually, and the oracle reading that triggers a claim is public before anybody calls settlement. Without a delay a depositor can read the trigger and exit ahead of settlement, leaving the loss with whoever stayed, which would make underwriting optional for the well-informed and compulsory for everyone else. A withdrawal request starts a cooldown, defaulting to three days and capped at fourteen, and the request snapshots the share balance held at request time so a pre-warmed empty address is worth nothing. Any outbound share transfer cancels the request. The cooldown in force is snapshotted with the request, so a later increase cannot retroactively freeze somebody who has already begun waiting.

Pausing never traps funds. Deposits are pausable and withdrawals are not. Opening a position is pausable; closing and liquidating are not. An emergency control that strands user collateral is not a safety mechanism.

11 Empirical validation

Every figure below was produced by running the repository's suites at the commit this paper describes, not quoted from documentation.

The suites that carry the claims in this paper are named rather than aggregated:

- `ReserveMonteCarloTest` produces Table 4 and Figure 10.
- `WaterfallMonteCarloTest` produces Figure 9.
- `PerilAggregationTest`, 19 tests, covers the correlated exposure caps including the requirement that a balancing trade is accepted over the cap and that a broken registry does not block trading.
- `PerpCapacityCapTest`, 14 tests, covers preventive capacity.
- `FirstLossIsolationTest`, 12 tests, covers the separation of tier 1 from the liquidation backstop.

Measure	Result
Test suites	54
Tests passing	568 of 568
Tests failing	0
Wall-clock runtime	114.9 s
Solidity contracts	29
Invariant runs	256 runs at depth 64
Fuzz iterations	10,000 per property
Priced strikes in the seeded climatology	1,080
Years of observational record per strike	30

Table 6: Suite results, measured. The full command is `forge test --summary` from the `contracts` directory.

- `PerpFundingScaleTest`, 29 tests, is the regression suite for the oracle scale defect.
- `NightmareScenariosTest` covers named worst cases, including the bank-run path that exposed the interaction between senior withdrawals and credited junior backing.

The invariants under continuous test are that market collateral plus insurance reserves cover aggregate open position equity; that the constant product k is preserved across every open and close; that cumulative funding payments balance exactly between longs and shorts; that long and short payouts sum to notional in every payoff mode; and that no path creates value or permanently strands funds.

12 Limitations and status

This section is deliberately specific. A whitepaper that does not state what is missing is not describing a system, it is advertising one.

The only functioning weather oracle is a mock. The FTSO and FDC adapters exist and satisfy the oracle interface, but they revert on read. Settlement is trusted input regardless of how the protocol is deployed. This is the largest single gap between the repository and a production system, and no amount of capital modelling compensates for it.

No mainnet deployment. The protocol is live on Flare Coston2 testnet only. The SDK, indexer, and frontend are genuinely parametrised by chain identifier rather than hardcoded, so a mainnet deployment needs a registry entry rather than a rewrite, but no mainnet deployment exists.

The published testnet addresses predate the capital structure. The live deployment is the pre-waterfall stack: no liquidity vault, junior tranche, first-loss reserve, peril registry, or policy market, and a single externally owned account as admin. Everything in Sections 6 through 9 exists in the repository and deploys from the deployment script, but is not at the published addresses.

The reserve parameter should probably move. Section 8 measured one payout shortfall at the shipped coverage ratio of 50% and none at 75%, for 0.48 percentage points of additional trade rejection. The failure boundary is also only located between 30% and 50% because 40% was never tested.

Linear and capped payoffs are unpriced. Fair-odds gating covers binary payoffs only. The others require the loss distribution rather than a single quantile of it, and are flagged unpriced on-chain rather than given an authoritative-looking wrong number.

Basis risk is not addressed. An index-settled contract pays against a measured index, not against the holder's actual loss, and the gap between the two is the buyer's [18]. The protocol makes the index explicit and settles it honestly; it does not reduce basis risk.

The oracle sampling cadence is trusted configuration. A market declares the interval at which its oracle publishes, and the protocol cannot verify it. Declaring a daily feed as weekly would under-count a monthly sum sevenfold. This is admin-only and bounded, and recorded as a disclosed limitation rather than pretended away.

No professional audit. Any mainnet deployment is gated on one.

13 Conclusion

The problem with on-chain weather markets has not been settlement. Parametric settlement against an oracle has been solved for years, and the protocols that do it well do it well. The problem has been that a hedger cannot transact until a counterparty appears, and for weather that counterparty is structurally absent, because everyone exposed to the same weather wants the same side of the same contract at the same time.

BreezeSwap's answer is to make capital the counterparty and to price from the climate record rather than from arriving orders, so the market has a price before it has participants. The work that makes this more than a slogan is in the constraints: gross collateralisation of cover so a claim cannot come up short by the premium that paid for it, a coverage ratio derived by sweeping the parameter and reading the results, a subordinated tranche that only buys capacity when the capital behind it is genuinely new, exposure caps that treat a shared storm as one risk rather than several, and a sale window wide enough that a buyer cannot price against a forecast the pool cannot see.

The measurements in this paper support the design and, in one place, argue against its current configuration. Both are reported. What remains between this and a production system is a working oracle, an audit, and a redeployment, and none of those is hidden in a footnote.

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